Introduction To Python Programming Language

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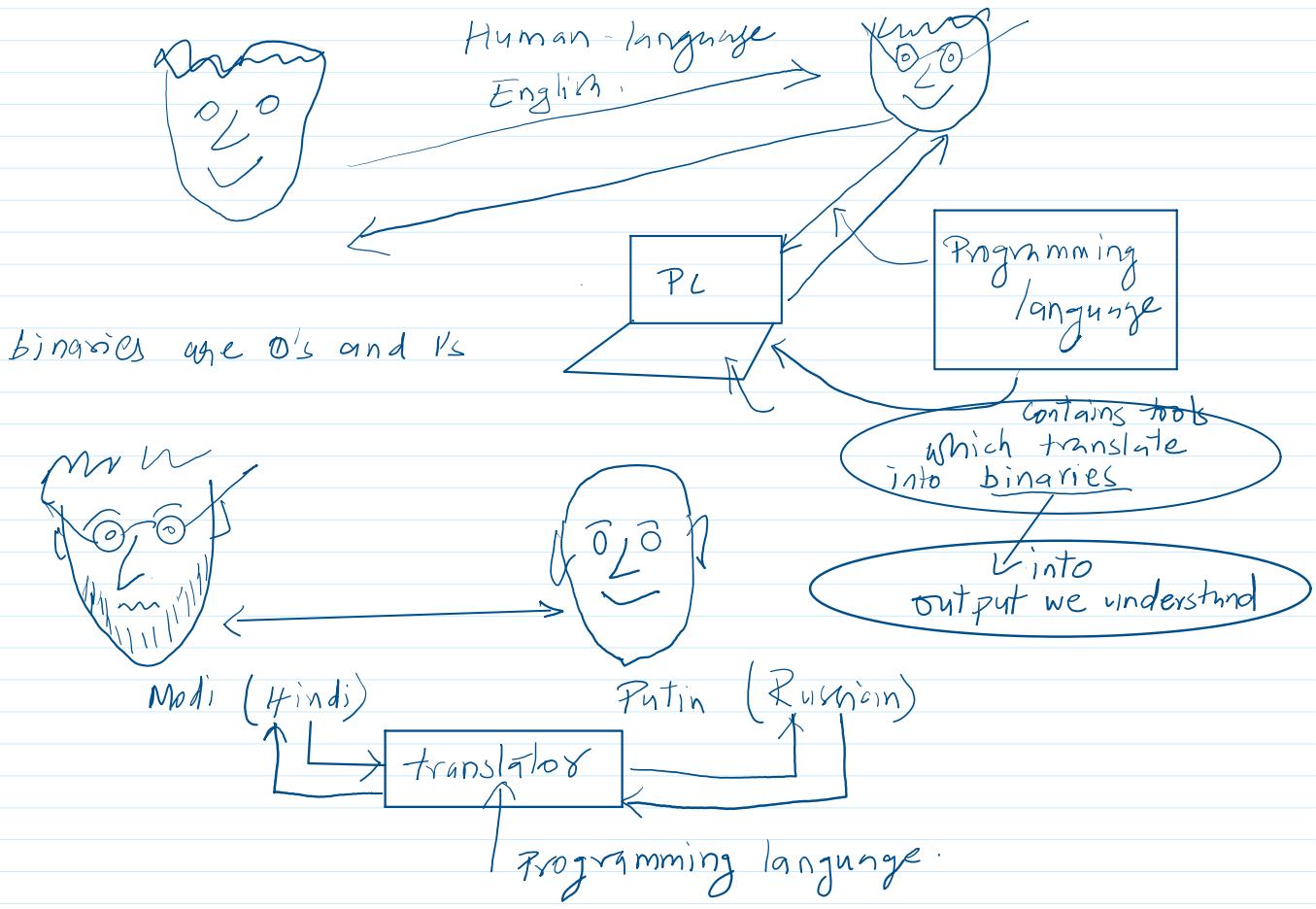
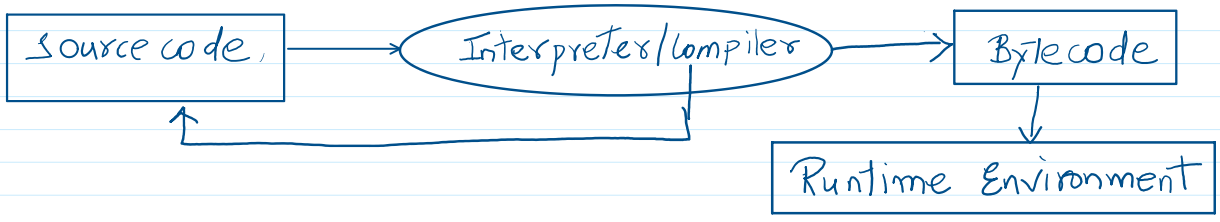


Writing programs in binaries is called low-level programming.
High-level programming language, which is close to Human language called "English"

A program written in high level language is called "source code".



Python Programming language

Python is a high level interpreted and general purpose programming language. Created by Guido van Rossum And first released in 1991. Python emphasizes code, readability and simplicity, making it an excellent choice for both beginners and experienced programmers.

Monty Python's Circus

Key feature of Python

1. **Simple and easy to learn:** Python's syntax is clear and intuitive resembling plain English, which makes it easy for newcomers to learn and understand.
2. **Interpreted language:** Python is an interpreted language, meaning that code is executed line by line, which facilitates quick testing and debugging.
3. **Dynamically typed:** Variables in python do not require an explicit declaration to reserve memory space. The declaration happens automatically when a value is assigned.
4. **High level language:** Python abstracts many of the complex details of the computer's operation, making it easier to focus on programming logic rather than hardware intricacies.
5. **Portable:** Python code can be run on various operating systems with little or no modification, ensuring cross platform compatibility.
6. **Extensive standard library:** Python has a rich standard library that supports many common programming tasks, such as file I/O, system calls and even Internet protocols.
7. **Object-oriented and functional programming:** Python supports both object oriented and functional programming paradigms providing flexibility in coding style.
8. **Extensible:** Python can be extended with modules written in C or C++, And it can be embedded into application to provide scripting capabilities.
9. **Community support:** Python has a large and active community that contributes to its vast ecosystem of libraries, frameworks and tool.
10. **Open source and free:** Python source code is also available to improve and it is free to all the developers

Using Python as Calculator

```
>>>7 % 4
   3
>>>7 // 4
   1
>>>7 / 4
   1.75
>>>4 ** 3
   64
>>>
```

remainder Operator

7 % 4 ← operation (Division)

output is Remainder

Operand

Quotient operator

7 // 4

4 ** 3 ← Exponential operator or to the power.